

ARYAMAN BHATNAGAR

Northampton, UK | 07466 179676 | aryaman.bhatnagar2900@gmail.com

Portfolio: <https://aryamanbhatnagarcad-portfolio.github.io/Aryaman-Bhatnagar/>

PERSONAL STATEMENT

Highly motivated and proactive Design Engineer specialising in mechanical design and lightweight structures, with an MSc in Automotive Engineering from Cranfield University. Experienced in CAD-driven product development applying DFM and DFA principles, with proven delivery of structural weight reduction through topology optimisation and FEA validation. Seeking early-career engineering opportunities in the automotive industry, focusing on innovative design, manufacturability, and advanced CAD proficiency in Siemens NX and PTC Creo.

TECHNICAL SKILLS

Core Competencies:

- **Mechanical Design:** Production Part Design, Parametric Modelling, Part Assembly, DFM, DFA, GD&T, Engineering Drawings, BOM, and Tolerance Stack-up Analysis
- **Automotive Systems Integration:** Powertrain packaging, installation constraints, subsystem compatibility, Electromechanical systems, and rotating components.
- **Structural Analysis:** Explicit FEA for Mechanical and Composite designs, Hand Calculation for validation, DFMEA, Fatigue analysis, Buckling, and Modal Analysis

Tools:

- **CAD:** CATIA (Intermediate), Siemens NX (Advanced), PTC Creo (Advanced), SolidWorks (Intermediate)
- **FEA:** ANSYS Mechanical (Advanced), HyperMesh (Intermediate), LS-DYNA (Beginner)
- **Others:** Team Centre, Windchill, MATLAB, and Simulink

KEY ACHIEVEMENTS

- 15% UAV mass reduction using topology optimisation, maintaining a FOS of 3.5 (AERO2ASTRO, 2021)
- 4.89% reduction in drag of a rider-motorcycle assembly through strategic placement of passive flow devices using Advanced CFD (Cranfield University, 2025)
- 25% weight reduction while maintaining a FOS of 3.2 (Cranfield University and McLaren, 2025)
- Conducted CAD and Engineering Drawing Workshops for 65+ students (Manipal University Jaipur, 2021)

CAREER HISTORY

Race Anywhere, Daventry, United Kingdom- Design Engineer

November 2025 - December 2025

The role ended due to management restructuring and the dissolution of the design team

- Reduced 35% mass of handbrake mount through topology optimisation with a FOS of 4 validated via ANSYS
- Designed production-ready mounts and applied DFM principles for CNC machining using SolidWorks
- Created engineering drawings and manufacturing documentation to ensure functional tolerances and seamless assembly compatibility
- Performed mechanical testing and QA validation of CNC-machined components, ensuring compliance with design tolerances.

Schindler Group, Mumbai, India- Executive- Quality Assurance

July 2022 - May 2024

- Reduced callbacks of the S3000 by 7% by conducting root cause analysis of mechanical components
- Improved the carbon emissions of maintenance vehicles by 3.84% by deploying 50+ electric vehicles
- Reduced ISO Audit non-conformities by 40% by conducting a stringent internal audit and root cause analysis

AERO2ASTRO, Tamil Nadu, India- Design Engineering Intern

November 2020 - April 2021

- Achieved 15% UAV mass reduction using topology optimisation and maintained a FOS of 3.5, validated through structural analysis
- Designed a UAV wing locking system in CAD, applying DFM principles for manufacturability using Siemens NX freeform surface modelling
- Developed a honeycomb sandwich structure, improving stiffness-to-weight ratio and validating performance using FEA

Infi League Motorsport, New Delhi, India- Chassis Design Engineering Intern August 2020 - September 2020

- Designed an off-road ATV chassis and composite monocoque using Generative Shape Design in CATIA V5
- Contributed to race-car systems integration by ensuring chassis design compatibility with suspension, drivetrain mounting points, and power unit installation interfaces in motorsport packaging constraints
- Implemented DFM strategies by defining carbon fibre laminate orientations and stacking sequences
- Achieved 18kg mass reduction using topology optimisation while maintaining torsional stiffness

ACADEMIC PROJECTS

MSc Automotive Engineering Dissertation: Using CFD to investigate the effect of triangular vortex generators on the aerodynamic performance of a naked-type motorcycle May 2025 - September 2025

- Developed a parametric CAD model of a motorcycle and rider geometry using Fusion 360 and Solidworks
- Conducted CFD simulations in ANSYS Fluent using the k- ω SST turbulence model
- Achieved 4.89% drag reduction through optimised vortex generator placement and iterative CFD simulations

MSc Automotive Engineering Group Design Project February 2025 - April 2025

- Designed a retrofittable hydrogen hybrid internal combustion engine system with integrated powertrain packaging considerations and installation constraints
- Designed a hydrogen fuel delivery system in PTC Creo, integrating pressure relief devices, non-return valves, shut-off systems, and thermal management components to meet hydrogen safety standards
- Executed DFMEA risk assessments, ensuring safety compliance, with only a 3.4% peak torque penalty
- Developed manufacturing-aware design concepts, including tooling and fixture considerations for assembly and installation feasibility in constrained engine bay environments

Vehicle Structures Module in collaboration with McLaren Automotive January 2025 - February 2025

- Redesigned front subframe using Creo and validated using HyperMesh FEA
- Performed fatigue, modal, and structural analysis under cyclic loading conditions
- Achieved 25% weight reduction with a safety factor of 3.2

EDUCATION

MSc in Automotive Engineering, Cranfield University, Cranfield, UK October 2024 - September 2025

- **Key Modules:** Vehicle Design, Vehicle Structures and Composites, Material and Manufacturing

B.Tech Mechanical Engineering, Manipal University, Jaipur, Rajasthan, India June 2018 - July 2022

- **Key Modules:** Solid Mechanics, Material Science, Machine Design, Thermodynamics, and Fluid Mechanics

INTERESTS, AND EXTRACURRICULAR ACTIVITIES

- **Languages:** English (Fluent), Hindi (Native)
- **Right to Work:** Graduate visa (No sponsorship required till December 2027)
- **Interests:** Track recent developments across Aerospace, Automotive and Motorsports, Swimming, Watching Formula 1, Hiking, Travelling
- **Extracurricular Activities:** Cranfield Walking Society: Regularly contributed to 20 km hikes, such as Chiltern Hills and Stewartby Lake, to promote fitness, teamwork, and appreciation of nature
- **Volunteering:** Contributed to raising awareness among 400 students in a school about road safety through murals, posters, and slogans for drivers to show how to drive safely